East African Cross-Border Trade:Finding Suspicious Volume Drops That Signal Smuggling or Conflict Summary

Introduction:

Analysing trade dynamics using FEWS NET Cross Border Trade Kenya, Uganda, Somalia

This project investigates which border crossing points show anomalous trade volume drops that correlate with known conflict events or policy changes, revealing the informal trade network's vulnerability points.

Since cross border trade in East Africa is highly sensitive to climate, policy and conflict, this analysis helps a lot.

Data Inspection:

Dataset Overview

Based on the data exploration in my notebook, all the datasets(f1,f2,f3) share identical structure of 46 columns. This is effective news because it means i can concatenate them into one master DataFrame easily.

Dataset1 (f1) has 21390 total rows
Dataset1 (f2) has 20162 total rows
Dataset1 (f3) has 39940 total rows

And since all three datasets have the same columns, concatenated into a total of 81, 492 rows of one combined dataset.

Checking null values, We have columns with null values:

Geography: reporting_country, border_point, source, destination
Product: product, cpcv2
Time : period_date, start_date
Metrics: value(primary trade volume/price),unit, common-unit_quantity, etc.

And we have columns with massive null values

Columns which indicate historical gaps such as value_five_years_ago,five_year_average,value_one_month_ago,value_one_year_ago and others are 90-95% empty.

Start and period_date dtype are objects, and there is a noisy list of units like 10_k, liquid measures(L,200_L) alongside standard metric tons(t).

ea(each) livestock counts or items.

No duplicate values found on the dataset.

Data Cleaning And Harmonization:

The raw datasets from FEWS NET were first inspected to understand their structure ,consistency, and data quality. All three dataset shared columns concatenated into one dataset.

Handling missing values:

A missing value revealed that several historical references such as value_five_years_ago, five_year_average contained almost 90% null values due to the high sparsity and limited analytical usefulness these columns dropped from the dataset to reduce noise.

Date Standardization:
The start_date and period_date columns were originally stored as objects, and they are converted into datetime format. Because they are very important for time series analysis.

Unit Standardization:
The dataset contains a wide variety of unit formats e.g 50_kg, 100_L, t,ea
Which required harmonization. All trade quantities standardized into metric tons(t) to ensure comparability across countries and commodities

Count based units(ea) typically representing livestock, separated from weight based analysis to avoid distortion.

Feature Engineering:
Week: weekly time index drive from period_date for time series aggregation cleaned product name to ensure consistency across datasets.
Quantity_mt: standardized trade volume in metric tons.

The dataset sorted by border point and time to ensure proper chronological order for time series computations such as percentage changes and anomaly detection

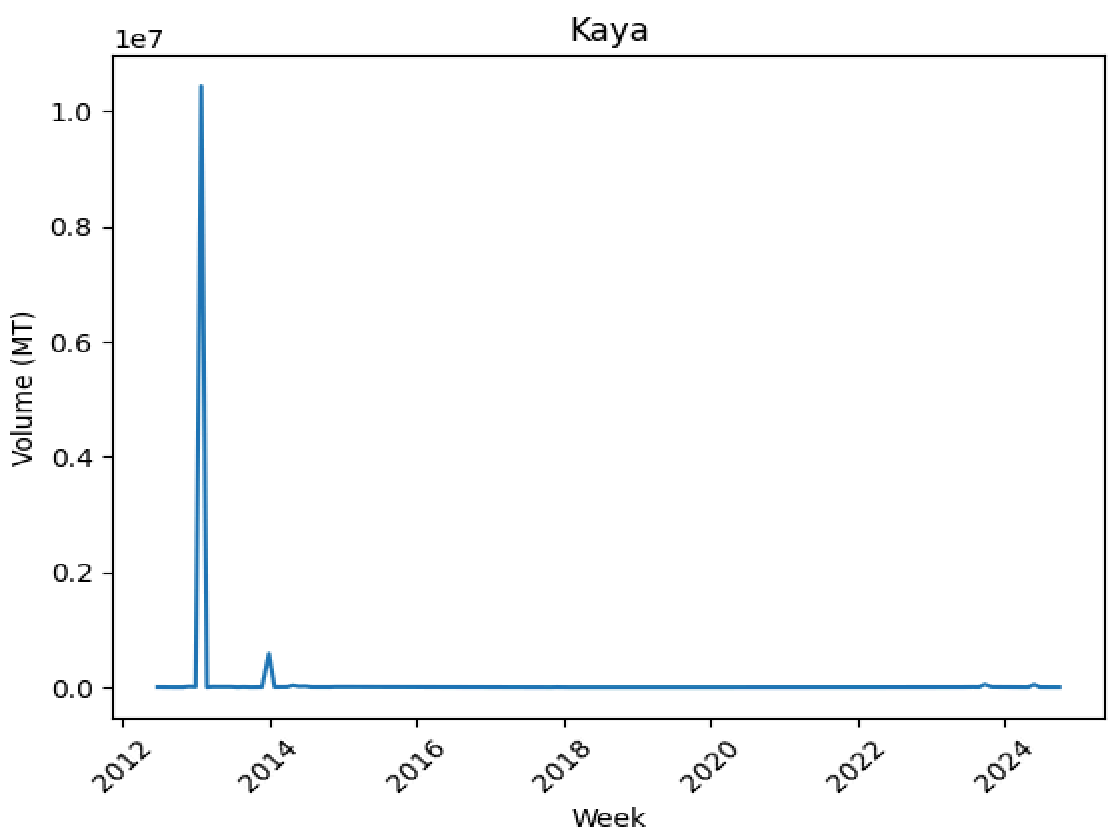
The analysis identified several high volume border crossings including key trade corridors such as Nimule, Kaya, Busia, Gutana and Feer-feer. These border points exhibited varying level of trade stability, with some showing consistent flows while others experienced significant volatility.

This analysis helps to identify anomalous volume drops and spikes that may signal disruptions such as conflict, policy changes or shifts in informal trade networks.

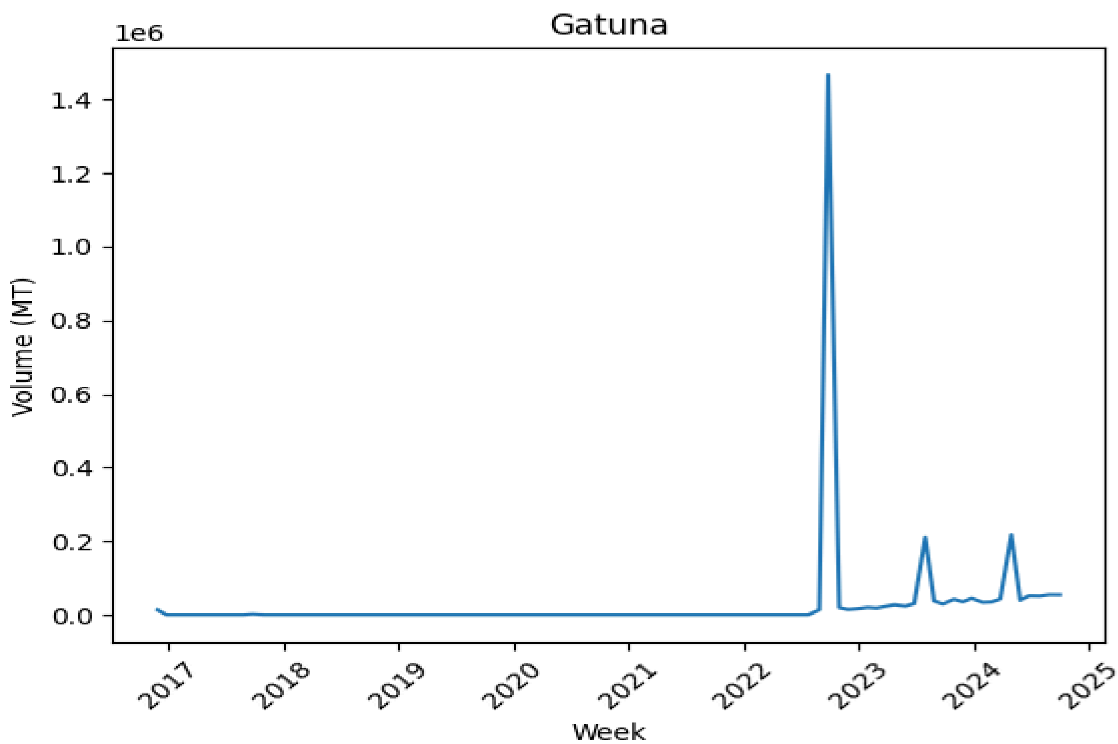
By applying time series analysis and anomaly detection techniques several border crossings were identified as key indicators of regional trade dynamics

Border Specific Trade Patterns:
The top five border crossing based on trade behavior:
Outlier -driven borders(Kaya, Gutana)

Kaya: shows a massive spike around 2013, reaching unusual high volumes that are not repeated. This suggests either data inconsistency or a one time large scale event such as humanitarian aid inflows.

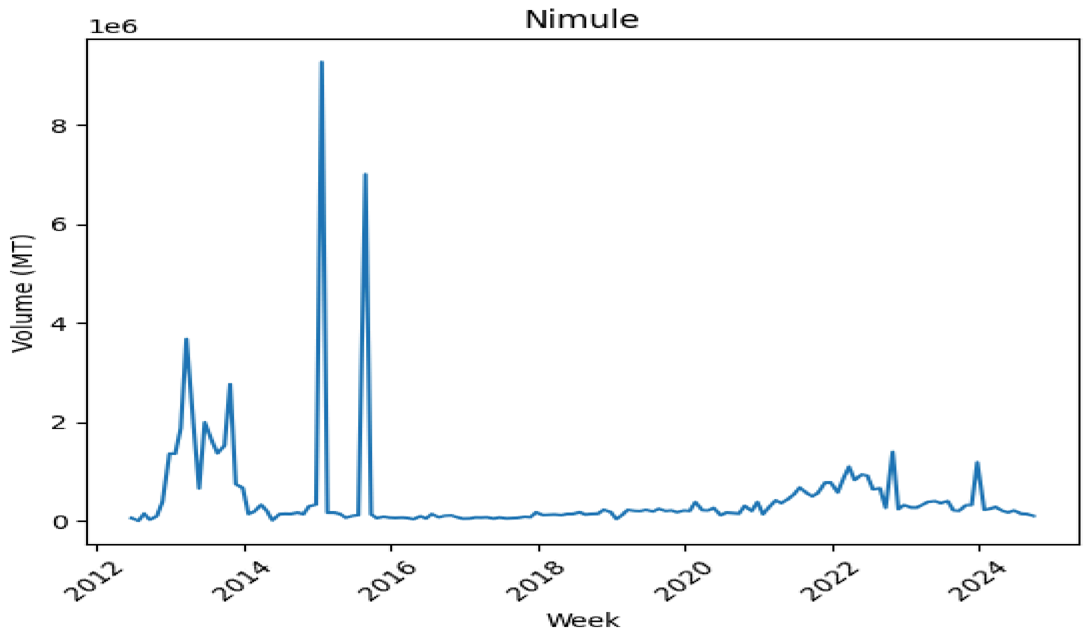


Gutana: exhibits a sharp surge in 2022 following a prolonged period of inactivity. This aligns with the reopening of the rwanda-uganda border after a multi year political closure.

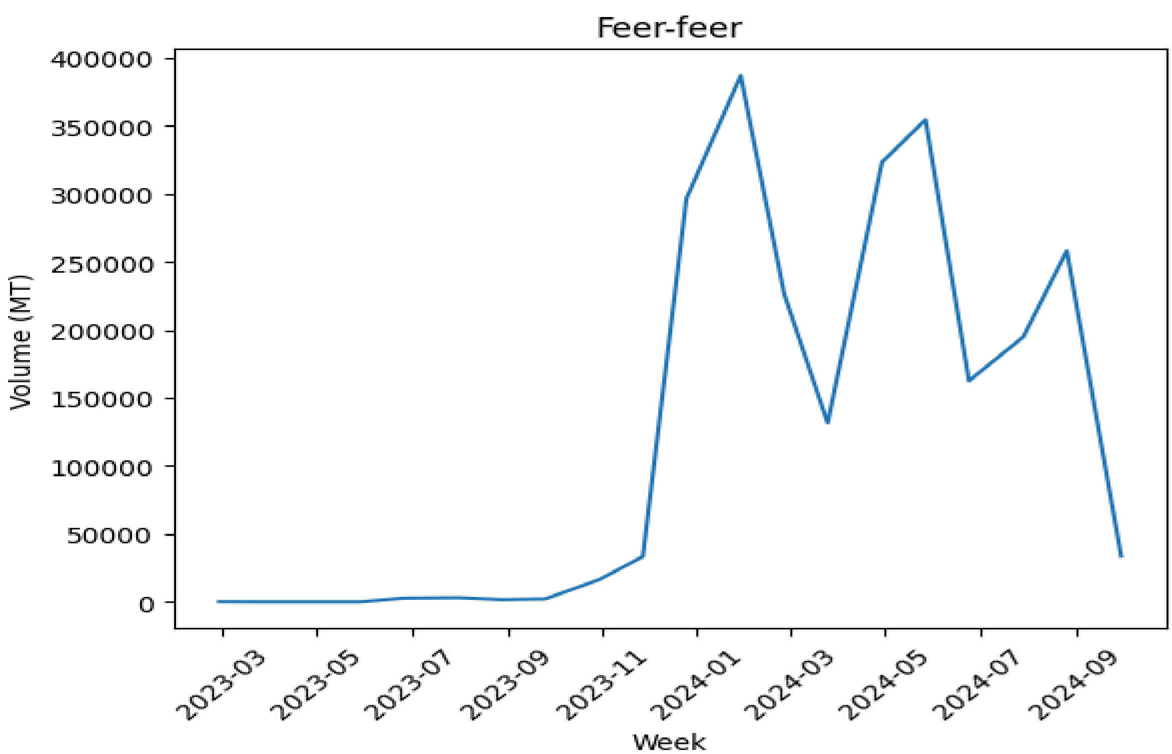


Nimule displays the longest and most consistent trade history, with activity dating back to 2012.
Large spikes observed between 2013 and 2016 likely correspond to humanitarian aid flows during periods of instability in South Sudan.

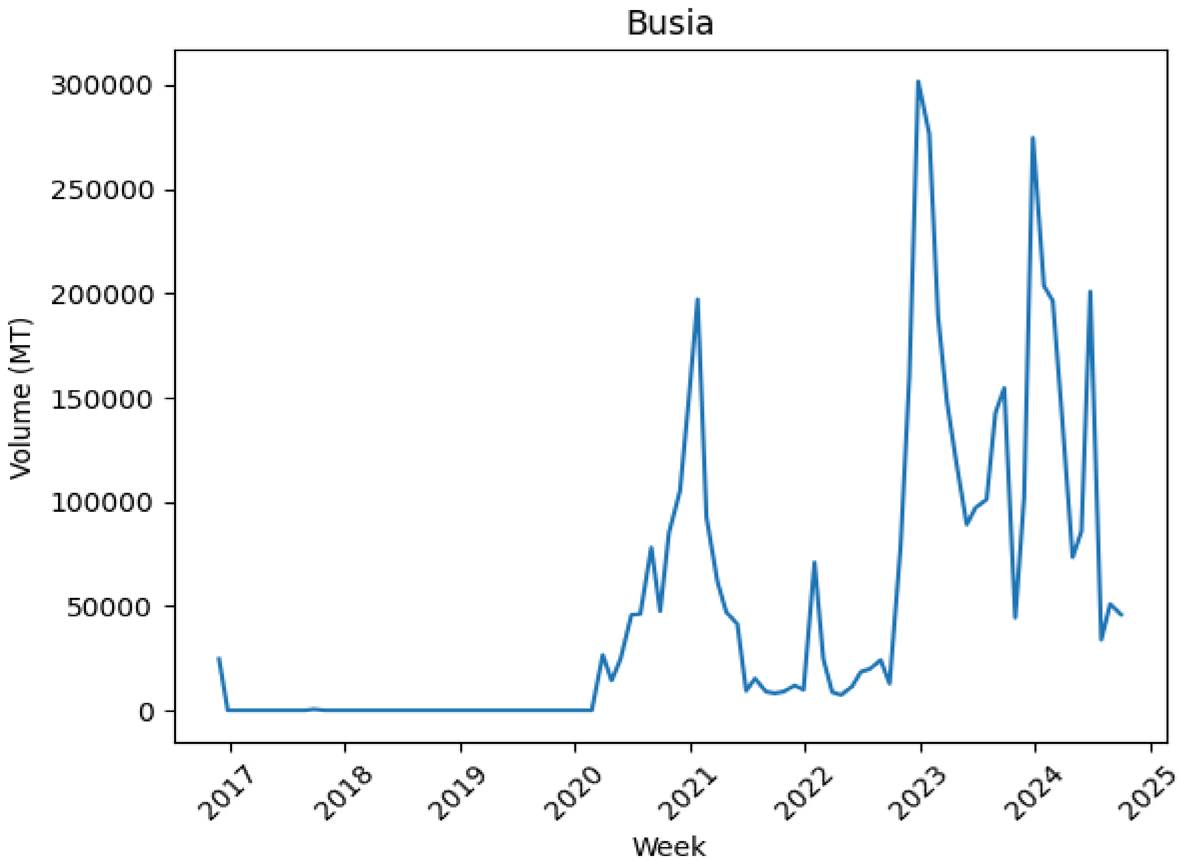
More recent years show reduced and and stabilized trade volumes, suggesting increased regulation or a shift in trade routes



Feer-Feer: demonstrates a rapid increase in trade activity beginning in late 2023. This suggests the emergence of a raw trade corridor, likely driven by improved security conditions and regional trade integration between Ethiopia and Somalia



Busia: shows high volatility, particularly in recent years, trade volumes fluctuate significantly week to week. Indicating sensitivity to logistics, customs processes and seasonal agricultural cycle.



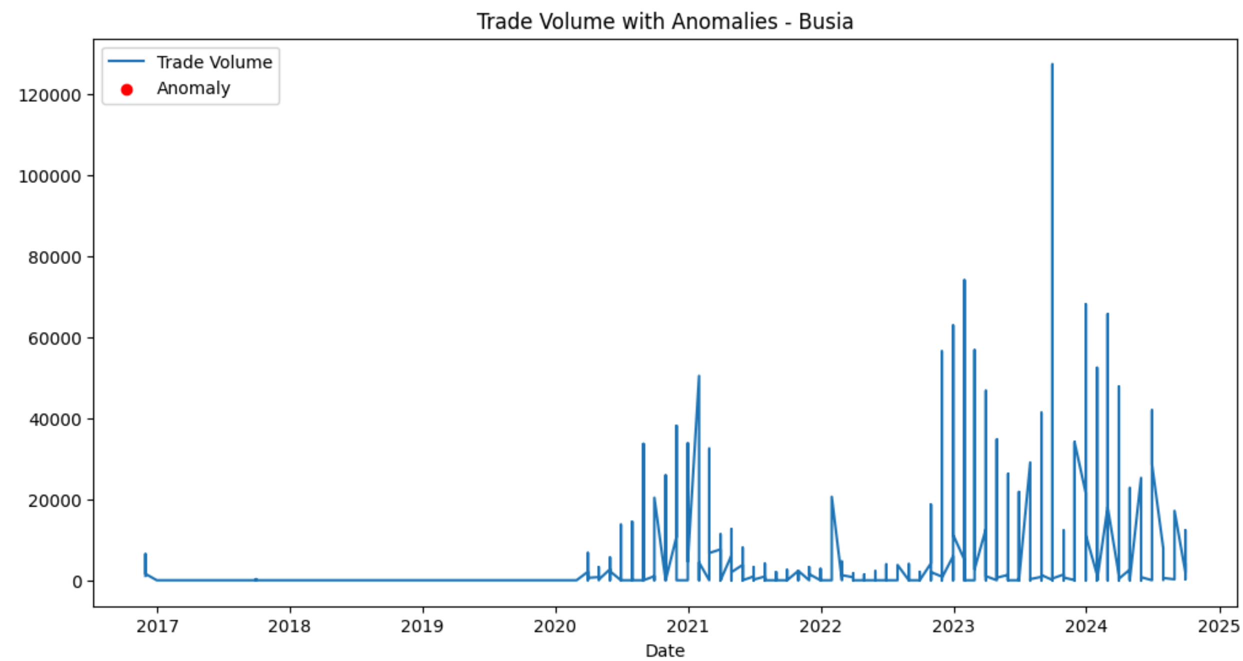
Time Series analysis revealed clear patterns of fluctuations in trade volumes Several border points demonstrated sharp spikes and drops, indicating periods of intensified or disrupted trade activity. These fluctuations were further investigated using anomaly detection.

Across multiple borders a noticeable decline or stagnation in trade volume occurs starting in early 2020. Border closures and strict health protocols slowed cross border movement. Considering covid19 disruption(2020-2021)

Spikes in Nimule and Kaya align with periods of instability in south Sudan . During conflict commercial trade decreases. These anomalies highlight the dual nature of trade data reflecting both economies and humanitarian aid flows.

When we see post conflict trade expansion, we can see Feer-feer, the surge in feer-feer in late 2023 shows trade expansion is likely driven by increased accessibility and reduced security risks. This suggests the opening of new trade opportunities following conflict resolution.

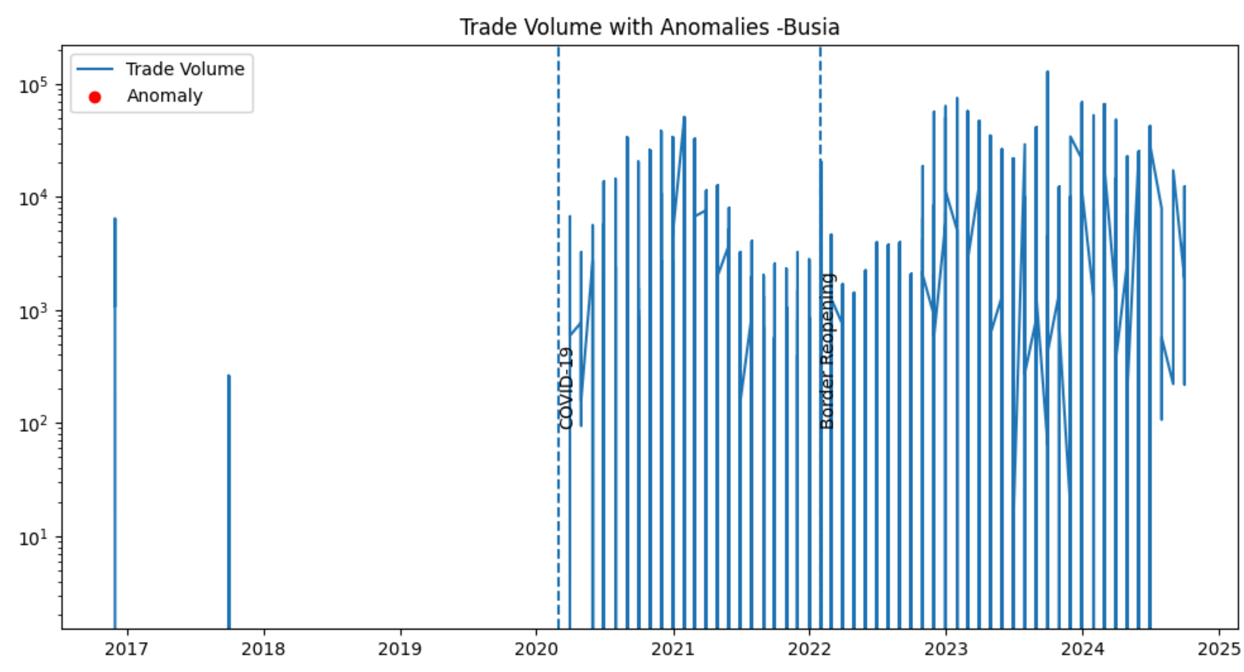
Using the zscore to detect anomalies when we check each border's total anomaly it shows 0 anomalies. For example, border_point Busia as we see on the following graph



The graph shows no anomaly with trade volume.

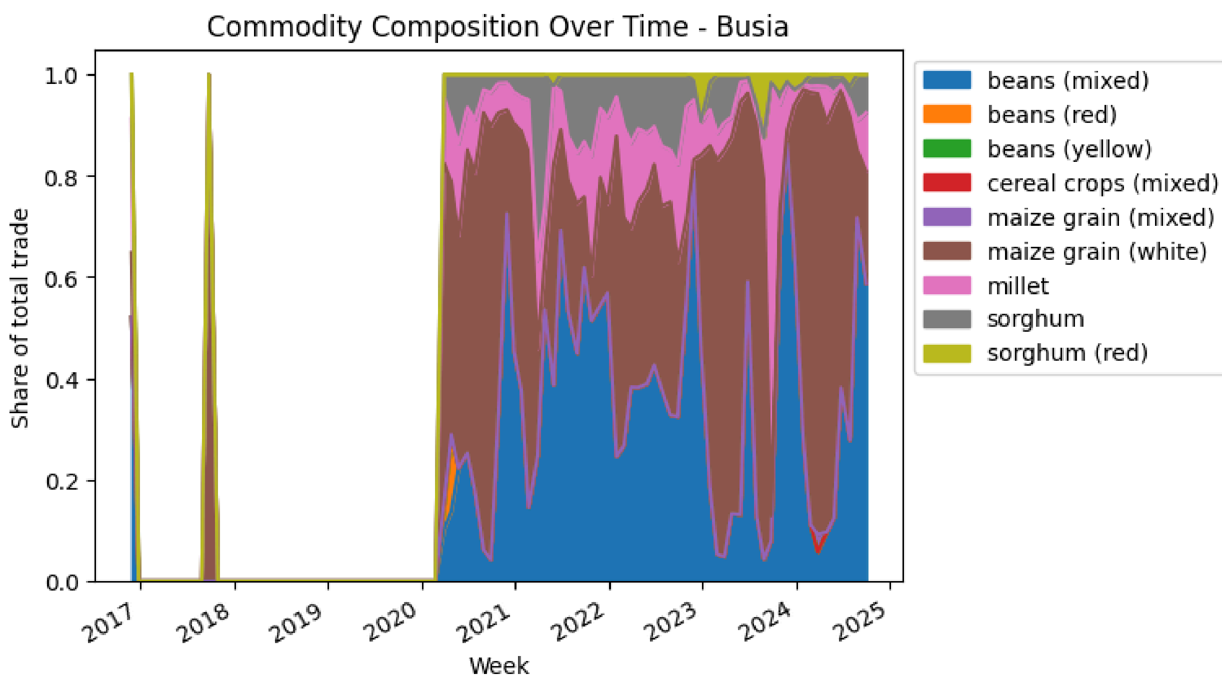
But we can see the change of the trade volume during some events like covid. As we see in the next picture.

At the time of covid, the border closed and we can see the trade volume when the border reopened.



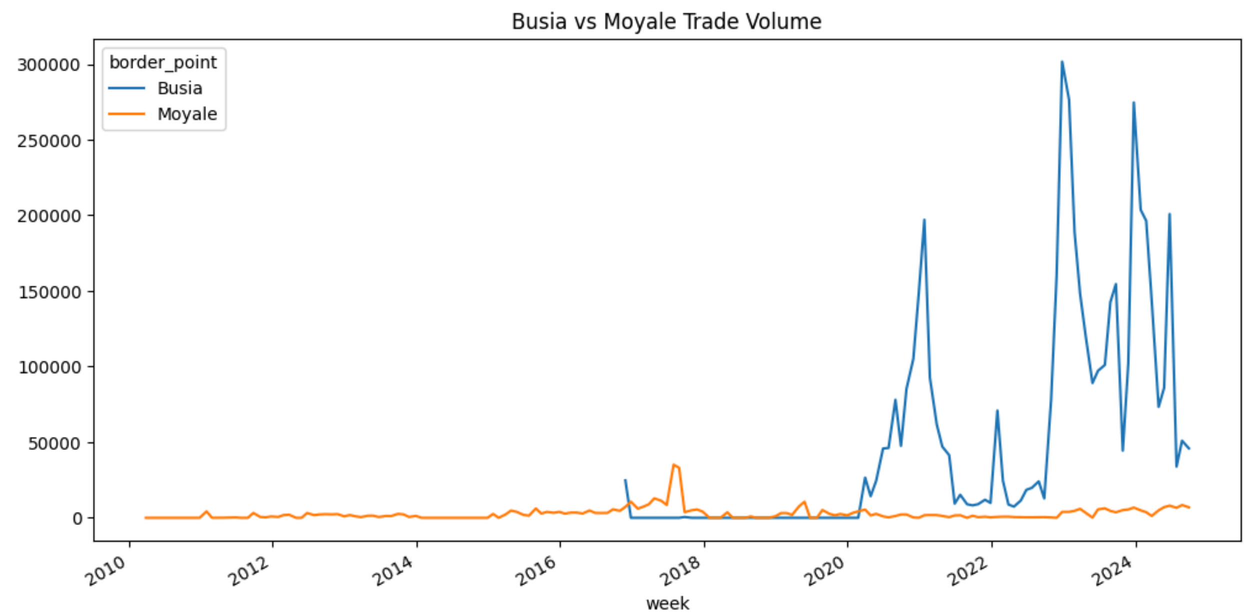
Commodity composition:

The commodity composition overtime shows grains share trades especially after covid there are massive grain products almost on each border. Beans and Maize grain(white) have a higher trade share. We can see on the next graph.



Cross border Comparison: Correlation analysis shows no substitution effects between kenya- Uganda trade at Busia and Kenya-Somalia trade at Moyale. The correlation coefficient(-0.025) is effectively zero. Indicating that changes in trade volumes at one border do not predict or offset changes at the other. Thus declines at Busia are not compensated by increased activity at Moyale; each border operates within independent trade systems driven by distinct regional dynamics.

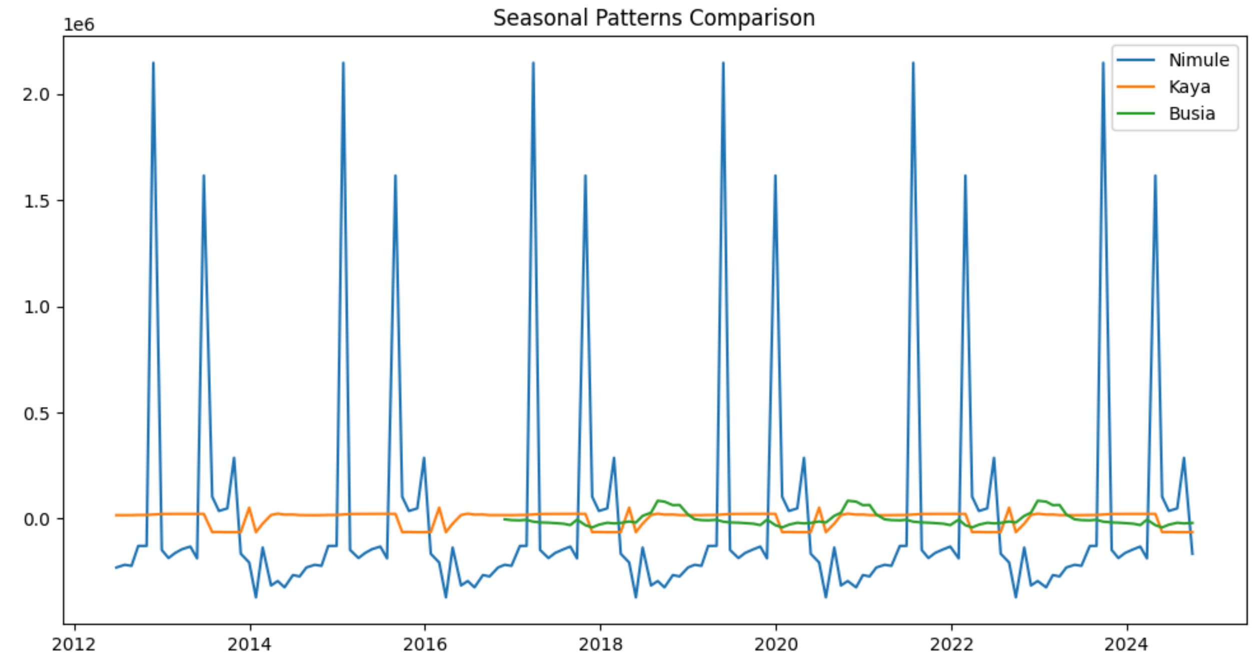
We can see on the following graph too.



Seasonal Decomposition:

Seasonal decomposition was performed using a 26 -week cycle due to limited data length (less than 104 observations required for annual seasonality).This adjustment allows for partial seasonal pattern analysis, capturing medium term fluctuations in trade activity.

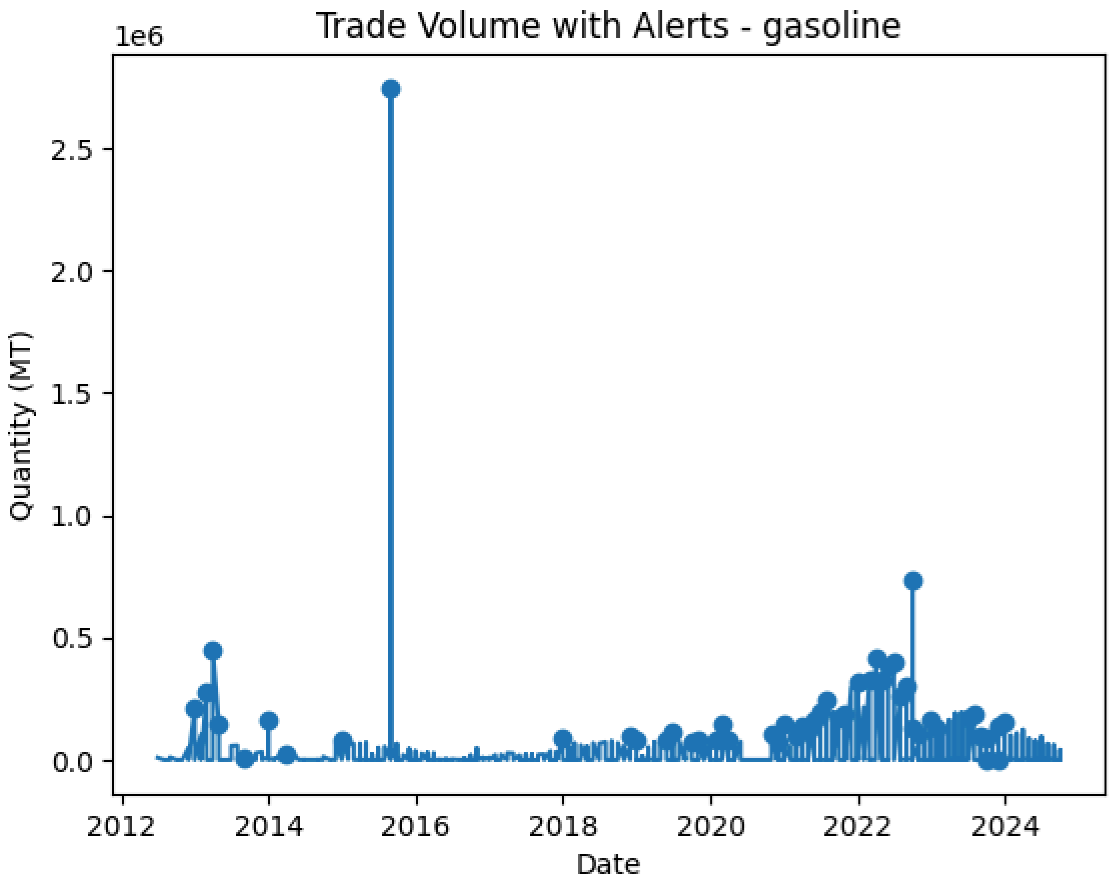
It reveals recurring annual patterns across key border points, consistent with regional agricultural cycles . Peaks in trade volume align with harvest season, while troughs correspond to lean periods indicating strong dependence on agricultural production cycle.

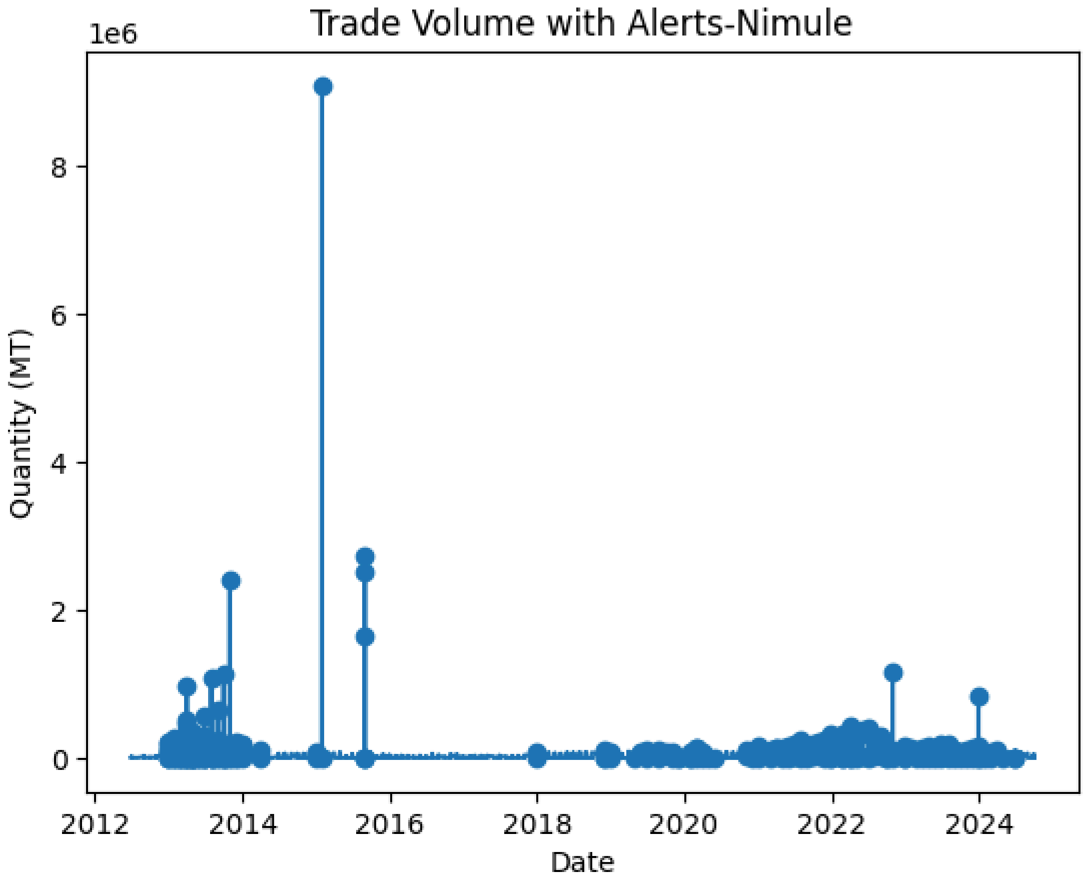


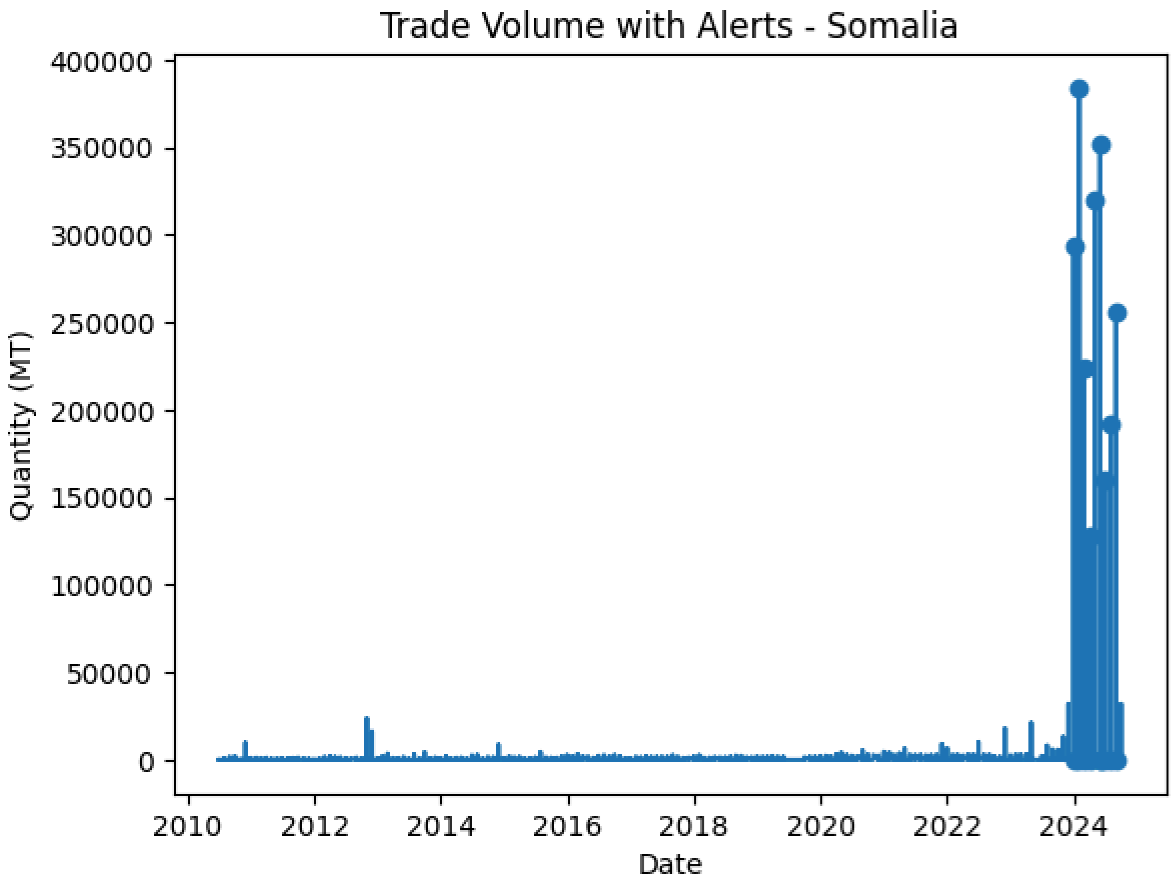
Alert System Insights:

A rule based alert system was implemented to detect suspicious trade disruptions. Alerts were triggered based on significant absolute changes in trade volume relative to historical variability.

This approach proved effective in capturing both sharp increases and decreases in trade activity


Trade Volume with Alerts - gasoline
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Quantity (MT)
2012
2014
2016
2018
2020
2022
2024
Date


Trade Volume with Alerts-Nimule
1e6
8
6
4
2
0
Quantity (MT)
2012
2014
2016
2018
2020
2022
2024
Date



The alert system successfully highlighted periods of abnormal behaviour across multiple border points, providing a useful tool for monitoring trade instability. However, the presence of zero values in the dataset introduced challenges for percentage based methods, requiring additional handling of infinite and undefined values

Implications and Recommendations

The findings suggest that certain border points are more vulnerable to trade disruptions and may serve as critical indicators of regional instability,monitoring these locations in real time could provide early warning signals for conflict, policy changes or supply chain disruptions.

It is recommended that: Trade monitoring systems incorporate anomaly detection techniques to identify unusual patterns early.

Border points with high volatility be prioritized for closer observation and intervention.

Data collection practices should be strengthened to reduce inconsistencies and improve reliability.

Additional contextual data(conflict events) be integrated to improve interpretation of anomalies

Conclusion

Overall, this project demonstrates the value of time series analysis and anomaly detection in understanding informal trade networks. By identifying patterns of disruption and instability, such Analysis can support better decision making for food security, trade policy, and regional stability in East Africa.